

Software Design Document

Workflow Queue (WFQ) Administrator Application

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Revision History

Name	Date	Reason For Changes	Version
Ivonne Arizpe	5/2/2026	Initial Document	1.0
Ruby Quinones	5/2/2026	Section 3	1.2
Ivonne Arizpe	5/3/2026	Section 1 Introduction, Section 4 System Architecture	1.2
Hector Corona	5/3/2026	Section 8: Database Design completion	1.3
Ivonne Arizpe	5/3/2026	Section 4, 6, and 7, 9	1.4
Janise Williams	5/4/2026	Section 9 User Interface	1.5
Ruby Quinones	5/6/2026	Section 10: Requirements Validation and Table of Contents	1.5
Ivonne Arizpe	5/6/2026	Section 12: References	1.5
Janise Williams	5/7/2026	Section 5: Policies and Tactics, Section 11: Glossary	1.5
Hector Corona	5/8/2026	Finalized Section 2, Removed API info	1.6
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Ruby Quinones	5/9/2026	Section 12.2 Regulatory References Revision	1.7
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Janise Williams	5/10/2026	Section 10 edited, removed API calling	1.8
Ivonne Arizpe	5/10/2026	Section 1 Introduction updated component descriptions	1.8
Ruby Quinones	5/10/2026	Removed Business Logic Mentions in 6.2	1.8
Janise Williams	5/11/2026	Revised section 6.4.6 and Section 4.1 DFD	1.8
Janise Williams	5/12/2026	Revised formatting for sections 7 and 8	1.8

1 Introduction

1.1 Purpose

This Software Design Document (SDD) describes the design decisions made by CS3338 Group 9 for the Workflow Queue (WFQ) Administrator Application. It explains how our team structured and planned the project, based on the ten tasks tracked in our Jira project board (FinalProject-Scrum). This document is intended to show how our design choices address the goals of each Jira task.

1.2 Scope

The WFQ Administrator Application is a web-based tool that allows authorized administrators to manage workflows, steps, and tasks. Our team's design covers the following areas, each derived directly from our Jira backlog:

- Database connectivity and error handling (FS-1, FS-9)
- Repository and version control setup (FS-2, FS-6)
- LaTeX documentation (FS-3, FS-4)
- Docker-based deployment (FS-5)
- TestRail integration for testing (FS-7)
- User authentication (FS-8)
- Bug tracking for login issues (FS-10)

1.3 Intended Audience

- **Group 9 Developers:** For implementation reference.
- **Course Instructor:** To evaluate the team's design decisions and task planning.
- **QA Testers:** To understand the system design when writing and executing test cases.

2 System Overview

The WFQ Administrator Application was planned and tracked using a Jira Scrum board with ten tasks (FS-1 through FS-10). Each task represents a component or area of work that our team designed and is responsible for. The system is built on ASP.NET Core MVC with a SQL Server database backend, containerized using Docker, and tested through TestRail.

The design of the system follows a layered approach:

- **Presentation Layer:** The web-based user interface served by ASP.NET Core MVC.

- **Business Logic Layer:** C# controllers that handle application logic and route requests.
- **Data Access Layer:** Entity Framework Core communicating with the SQL Server database.
- **Infrastructure Layer:** Docker containers for the frontend, backend, and database services.

3 Design Decisions by Jira Task

This section describes the design decision and approach for each task on our Jira board.

3.1 FS-1: Fix Database Connection

The database connection is configured through the application's `appsettings.json` file using a named connection string. The connection string points to the SQL Server container defined in our `docker-compose.yml`. Entity Framework Core is used as the ORM to manage all database interactions.

To make the connection reliable, the backend service in Docker Compose uses a `depends_on` directive to ensure the database container is running before the application starts. Connection credentials are stored in environment variables and are not hardcoded in the source code.

3.2 FS-2: Repository Setup

The GitHub repository is organized with the following folder structure:

- `/docs` — Contains all $\text{\LaTeX}$ documentation files including the SRS, SDD, Design Specification, Snapshot Objectives, and User Manual.
- `/testrail` — Contains TestRail test run reports for Snapshots 2, 3, and 4 in Markdown format.
- `README.md` — The project's user manual providing an overview, Jira link, and access instructions.
- `docker-compose.yml` — The Docker Compose configuration for running all services.

All team members were given collaborator access to the repository and committed work throughout the project lifecycle.

3.3 FS-3: $\text{\LaTeX}$ SRS (Version 1)

The Software Requirements Specification was written in $\text{\LaTeX}$ and stored at `/docs/SRS/SRS.tex`. It defines the functional and non-functional requirements for the WFQ application from the

perspective of our team’s planned work. The SRS was written by our team and is based on the expected behaviors described in our TestRail test cases.

3.4 FS-4: LaTeX SDD (Version 2)

This document (the SDD) is the deliverable for FS-4. It is written in $\text{\LaTeX}$ and stored at `/docs/SSD/SDD.tex`. It describes how the system was designed and how each Jira task maps to a design decision. The SDD is written entirely by our team and reflects our own project planning and design choices.

3.5 FS-5: Establish Docker

The Docker setup is defined in `docker-compose.yml` at the root of the repository. The configuration includes three services:

- **wfq-frontend** — A Node.js container serving the frontend of the application on port 3000.
- **wfq-admin-app** — An ASP.NET Core container running the backend application on port 8080.
- **wfq-database** — A Microsoft SQL Server 2019 container running on port 1433, with a persistent volume for data storage.

The frontend depends on the backend, and the backend depends on the database, ensuring services start in the correct order. Database credentials are passed through environment variables.

3.6 FS-6: Start Git Repo

The GitHub repository was initialized at the start of the project with an initial commit containing the folder structure and README. The main branch serves as the primary branch for all finalized work. Team members worked on updates and committed directly to main for finalized changes. The repository is hosted at: <https://github.com>

3.7 FS-7: TestRail Connection

TestRail was used to manage and execute all test cases for the project. Test runs were created for Snapshots 2, 3, and 4. After each test run, a summary report was exported and saved in the `/testrail` directory of the repository. The TestRail reports include test case titles, expected results, and Pass/Fail status for each test.

3.8 FS-8: Authentication

User authentication is handled by the application before granting access to any administrative page. The design uses a URL query string parameter (`?is_admin=true`) to determine admin authorization, which is checked on every request. If a user is not authorized, they are

redirected to an Access Denied page. This approach was chosen for simplicity within the scope of the course project.

3.9 FS-9: SQL Error

SQL error handling is implemented at the data access layer. All database calls are wrapped in try-catch blocks that catch `SQLException` and general `Exception` types. When an error is caught, the system logs the details to the application log and displays a user-friendly error message on screen. The application does not crash or expose raw SQL error details to the end user.

3.10 FS-10: Potential Bug — User Login Issues with Credentials

This Jira item was created as a bug ticket to track a known area of weakness: user login may fail when certain types of credentials are entered, such as those containing special characters or empty fields. This bug was logged with Highest priority because login is a critical entry point for the system.

The design response to this bug is to ensure that the authentication layer validates and sanitizes all credential inputs before processing. Error messages shown to the user must be informative but must not reveal system-level details.

4 Technology Stack

The following technologies were selected by our team for this project:

Technology	Purpose
ASP.NET Core MVC	Web application framework for the backend and UI
C#	Primary programming language
Entity Framework	ORM for database access
SQL Server 2019	Relational database for data storage
Docker	Containerization of all services
Docker Compose	Orchestration of multi-container deployment
GitHub	Version control and collaboration
Jira	Task and bug tracking
TestRail	Test case management and execution
L ^A T _E X	Documentation formatting

5 Constraints and Limitations

- The application is a course project and is not deployed to a live production environment.

- Authentication is simplified using a URL query string flag rather than a full identity provider integration.
- The database schema used is based on an existing project template provided as part of the course.
- The project scope is limited to what is trackable within the six-snapshot course timeline.

A Jira Task Summary

Jira ID	Task	Assignee
FS-1	Fix Database Connection	Hector Corona
FS-2	Repository Setup	Ruby Quinones
FS-3	LaTeX SRS (Version 1)	Ivonne Arizpe
FS-4	LaTeX SDD (Version 2)	Janise Williams
FS-5	Establish Docker	Janise Williams
FS-6	Start Git Repo	Hector Corona
FS-7	TestRail Connection	Ivonne Arizpe
FS-8	Authentication	Ruby Quinones
FS-9	SQL Error	Ruby Quinones
FS-10	Potential Bug: User Login Issues	Hector Corona

B References

- Jira Project Board: FinalProject-Scrum
<https://calstatela-team-uqomn8ag.atlassian.net/jira/software/projects/FS/boards/133>